

Atmosic ATM33/e brief summary

July 2023

Foundation for Product Generations

ATM2/ATM3 Series



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- More Processing & Embedded NVM
- Security and ARM TrustZone
- Bluetooth LE 5.1/5.2/5.3
- AoA/AoD for Asset Tracking, Direction Finding
- PV/RF Harvester and Battery Support Enhancements
- Sensor Hub Enhancements – MCU-less Operation
- Increased Tx Power

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ATM33/e Series



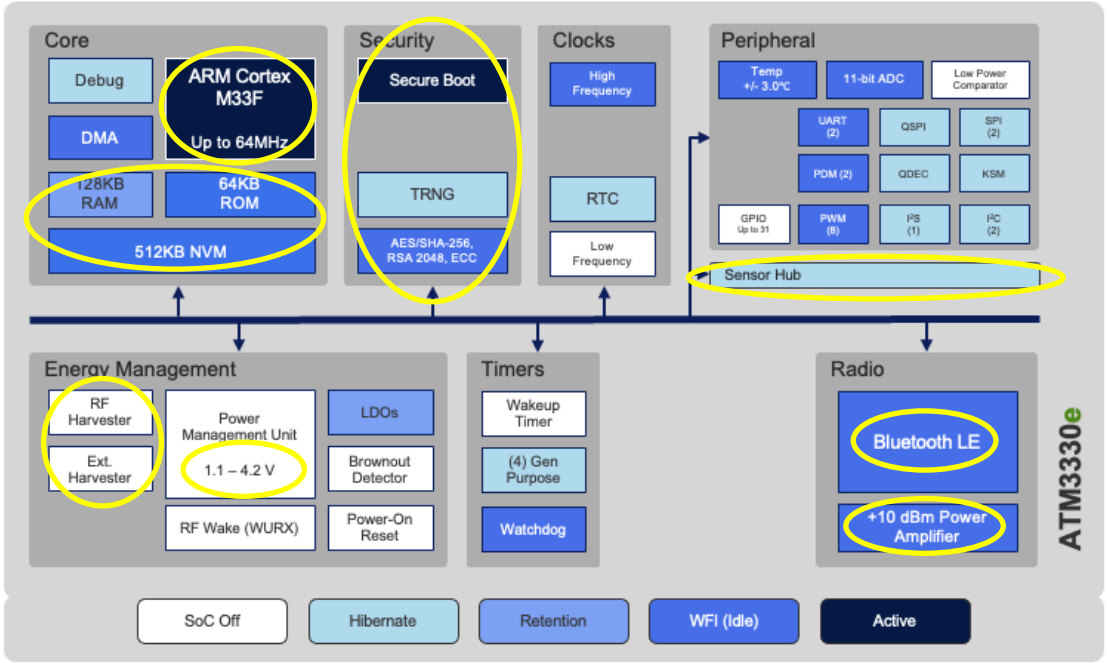
Common Foundation for All Product Generations –

- **Ultra Low Power Consumption**
- **Managed Energy Harvesting (PV, RF, Thermal, Mechanical)**
- **MCU-less Sensing, On-Demand Wakeup Receive (WuRx)**

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ATM33/e Series Product Features

- Bluetooth LE 5.3
- Radio power consumption
  - Receiving: 0.85 mA@-95dBm
  - Transmitting: 2.1 mA@0dBm
- RX Sensitivity: -95dBm
- TX Output: +10dBm
- Energy Harvesting (ATM33e only)
- ARM M33F @ 64MHz
- ARM TrustZone, Secure Boot & Secure Debug, Secure OTA
- TRNG, AES-256 HW Crypto
- NVM 512kB, RAM 128kB, ROM 64kB
- QFN 5x5mm, 7x7mm, WLCSP



ATM3330e - Energy Harvesting
ATM3330 – non Energy Harvesting

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ATM33/e Chip Packages

P/N	Package	Energy Harvesting	GPIO	MP
ATM3330e	7x7 QFN, 56 pins	YES	31	Now
ATM3330	7x7 QFN, 56 pins	NO	31	Now
ATM3325	5x5 QFN, 40 pins	NO	21	Now
ATM3325	WLCSP, 49 pins 2.610mm x 2.430mm x 0.377mm	NO	21	Q4 23' (ES on 9/30)

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Atmosic ATM33 vs. Competition

	Atmosic ATM33	NRF52840	NRF52833	NRF52832	TLSR8271	EFR32BG22
Radio-only RX Power (@3V)	**0.85 mA**	4.6mA	4.6mA	5.4mA	N/A	N/A
Radio-only TX Power (@3v @0dBm)	**2.1 mA**	4.8mA	4.9mA	5.3mA	N/A	N/A
SoC Rx Power (@3V)	**1.4 mA**	N/A	N/A	N/A	4.6mA	3.6mA
SoC Tx Power (@3v @0dBm)	**2.5 mA**	N/A	N/A	N/A	4.9mA	4.1mA
RX Sensitivity (1Mbps)	-95dBm	-95dBm	-96dBm	-96dBm	-96dBm	-98.9dBm
TX output	**10dBm**	8dBm	8dBm	4dBm	10dBm	6dBm
CPU	64Mhz M33F	64MHz M4F	64MHz M4F	64MHz M4F	48MHz RISC	76.8MHz M33

Atmosic Advantage:

- Extreme Low Power
- Advanced Computation
- Advanced Security
- Energy Harvesting
- On-demand Wake Up
- MCU-less Sensing

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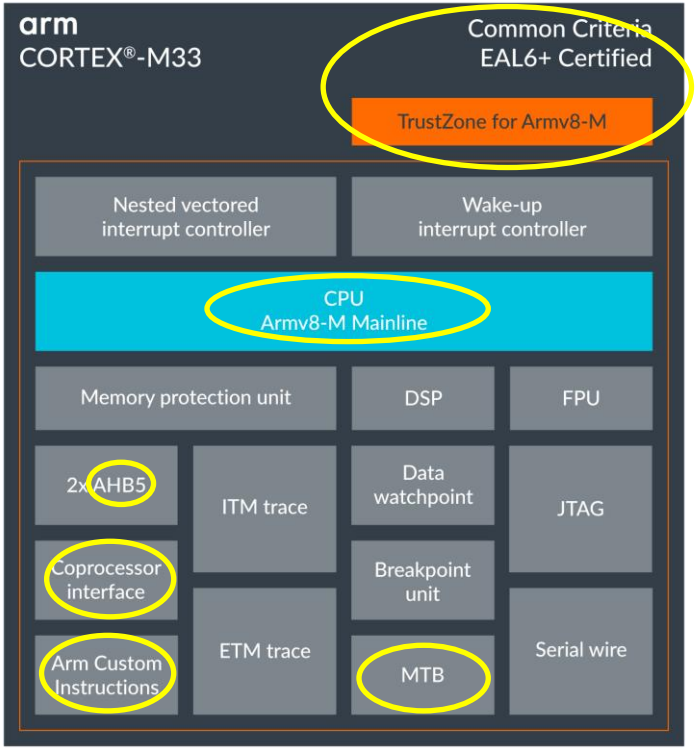
Atmosic ATM33 vs. Competition - continue

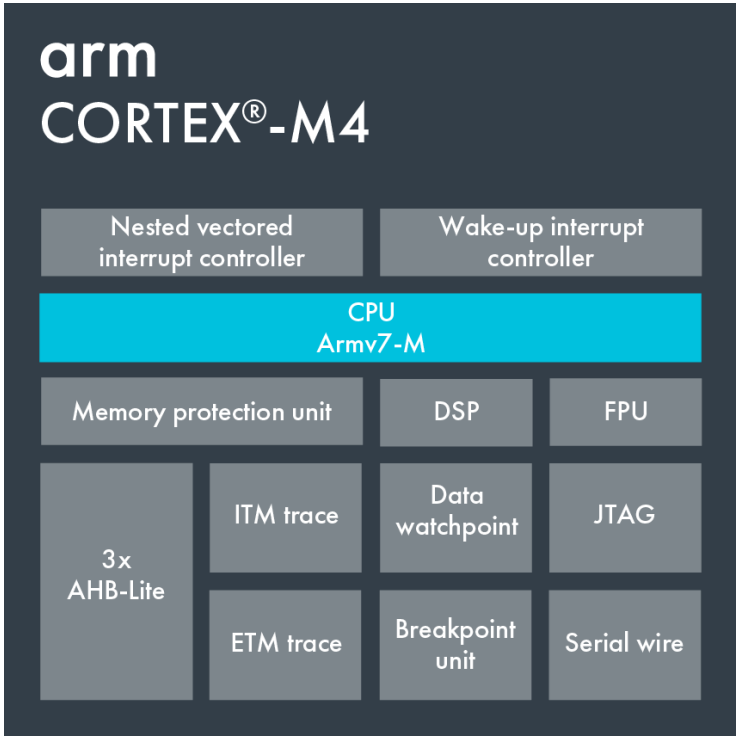
	Atmosic ATM33	NRF52840	NRF52833	NRF52832	TLSR8271	EFR32BG22
AoA, AoD	Yes	No	Yes	Yes	Yes	Yes
Voice Scaling	**Yes**	No	No	No	No	No
Voice Activity Detect	**Yes**	No	No	No	No	No
ARM TrustZone	**Yes**	No*	No	No	No	Yes
AES	**256(HW)**	128	128	128	128	256
NVM/SRAM	512KB/128KB	1MB/256KB	512KB/128KB	512KB/64KB	512KB/32KB	512KB/32KB
ROM	**64KB**	No	No	No	No	No
ADC	11b	12b	12b	12b	14b	16b
Mic	Digital	Digital	Digital	Digital	Analog, Digital	Digital
Voltage	1V-4.2V	1.7V - 5.5V	1.7V - 5.5V	1.7V - 3.6V	1.8V - 3.6V	1.71V - 3.8V
On Demand Wake Up	**Yes**	No	No	No	No	No
Energy Harvesting	**Yes**	No	No	No	No	No
Operating Temperature	-40C to 85C	-40C to 85C	-40C to 105C	-40C to 85C	-40C to 85C	-40C to 125C

*nRF52840 included ARM CryptoCell cryptographic accelerator, which only supply the crypto toolbox functionality as an IP block (e.g. HW accelerated cryptography and secure key derivation, for Secure Boot with Root-of-Trust). **nRF52840 does not provide full *Trusted Execution Environment* as described by ARM TrustZone.**

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ARM M33F vs. M4 – Block Diagrams





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Thank you